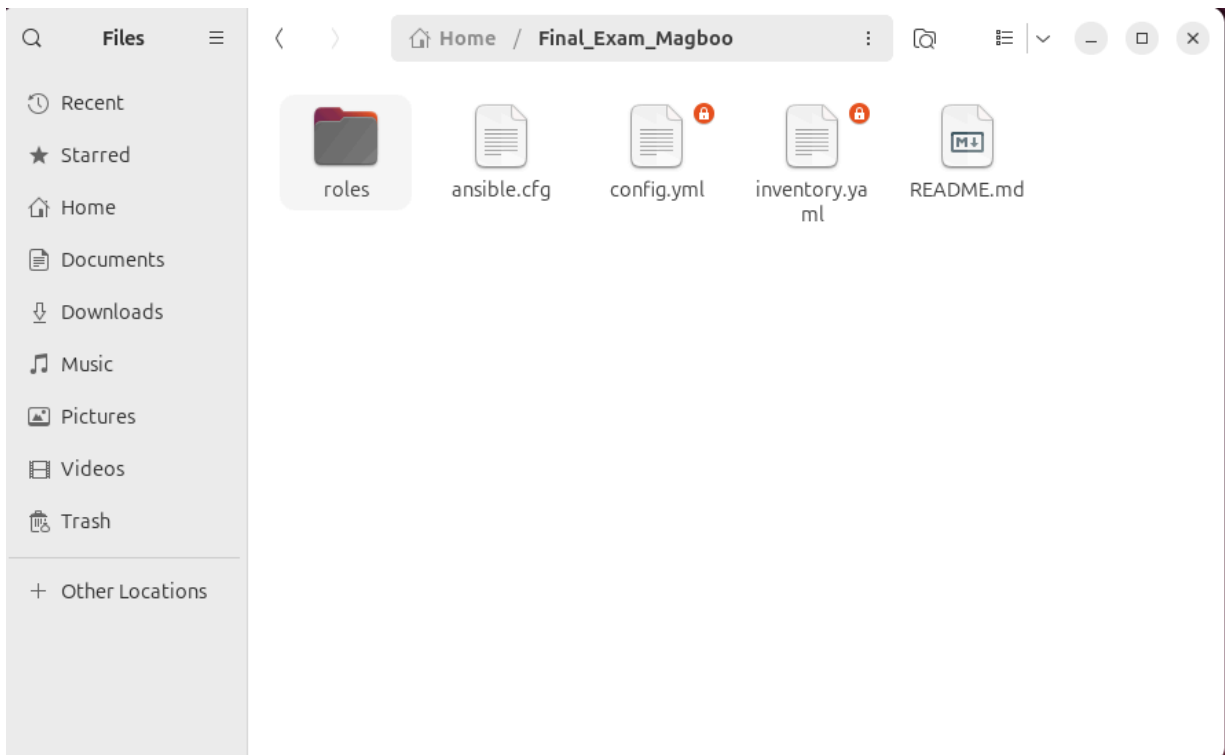


Hands-on Final Exam																
Name: Magboo, Matt Clemence C.	Date: 11/14/2025															
Section/Course Code: CPE31S2/CPE212	Instructor: Engr. Robin Valenzuela															
Tools Needed:																
1. VM with Ubuntu, CentOS and Ansible installed																
2. Web browser																
Procedure:																
1. Create a repository and label it as "Final_Exam_Surname"																
 Final_Exam_Magboo Public Pin Watch 0 main 1 Branch 0 Tags Go to file Add file Code  MagbooMattClemence Final Exam 7681661 · 4 minutes ago 3 Commits <table><tr><td>roles</td><td>Final Exam</td><td>4 minutes ago</td></tr><tr><td>README.md</td><td>Initial commit</td><td>5 days ago</td></tr><tr><td>ansible.cfg</td><td>Final exam part 1</td><td>5 days ago</td></tr><tr><td>config.yml</td><td>Final Exam</td><td>4 minutes ago</td></tr><tr><td>inventory.yaml</td><td>Final Exam</td><td>4 minutes ago</td></tr></table> README Final_Exam_Magboo		roles	Final Exam	4 minutes ago	README.md	Initial commit	5 days ago	ansible.cfg	Final exam part 1	5 days ago	config.yml	Final Exam	4 minutes ago	inventory.yaml	Final Exam	4 minutes ago
roles	Final Exam	4 minutes ago														
README.md	Initial commit	5 days ago														
ansible.cfg	Final exam part 1	5 days ago														
config.yml	Final Exam	4 minutes ago														
inventory.yaml	Final Exam	4 minutes ago														
2. Clone your new repository in your VM																



3. Create an Ansible playbook that does the following with an input of a config.yml file and structure inventory file.

```
Magboo@LocalMachine: ~/Final_Exam_Magboo
GNU nano 7.2 config.yml
--
- hosts: all
  become: true
  pre_tasks:

    - name: Disable epel-next on CentOS (prevents broken dependencies)
      command: dnf config-manager --set-disabled epel-next
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: Update packages on CentOS
      tags: always
      dnf:
        name: "*"
        state: latest
        exclude:
          - centos-release*
          - centos-stream-release*
          - system-release*
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: Update and Install Update (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        changed_when: false
        when: ansible_distribution == "Ubuntu"

[ Read 41 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-G Copy
```

```

- name: motd play
  # MESSAGE OF THE DAY
  copy:
    content: "Ansible Managed by Magboo\n"
    dest: /etc/motd

- hosts: all
  become: true
  roles:
    - prometheus
    - apache2

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```

Magboo@LocalMachine: ~/Final_Exam_Magboo

TASK [prometheus : install snapd for CentOS] *****
skipping: [Server1]
ok: [Server2]

TASK [prometheus : enable snapd socket] *****
skipping: [Server1]
ok: [Server2]

TASK [prometheus : ensure /snap symlink exists] *****
skipping: [Server1]
ok: [Server2]

TASK [prometheus : wait for snapd to initialize] *****
skipping: [Server1]

TASK [prometheus : install prometheus with snapd] *****
skipping: [Server1]
changed: [Server2]

TASK [apache2 : install apache and php for Ubuntu] *****
skipping: [Server2]
ok: [Server1]

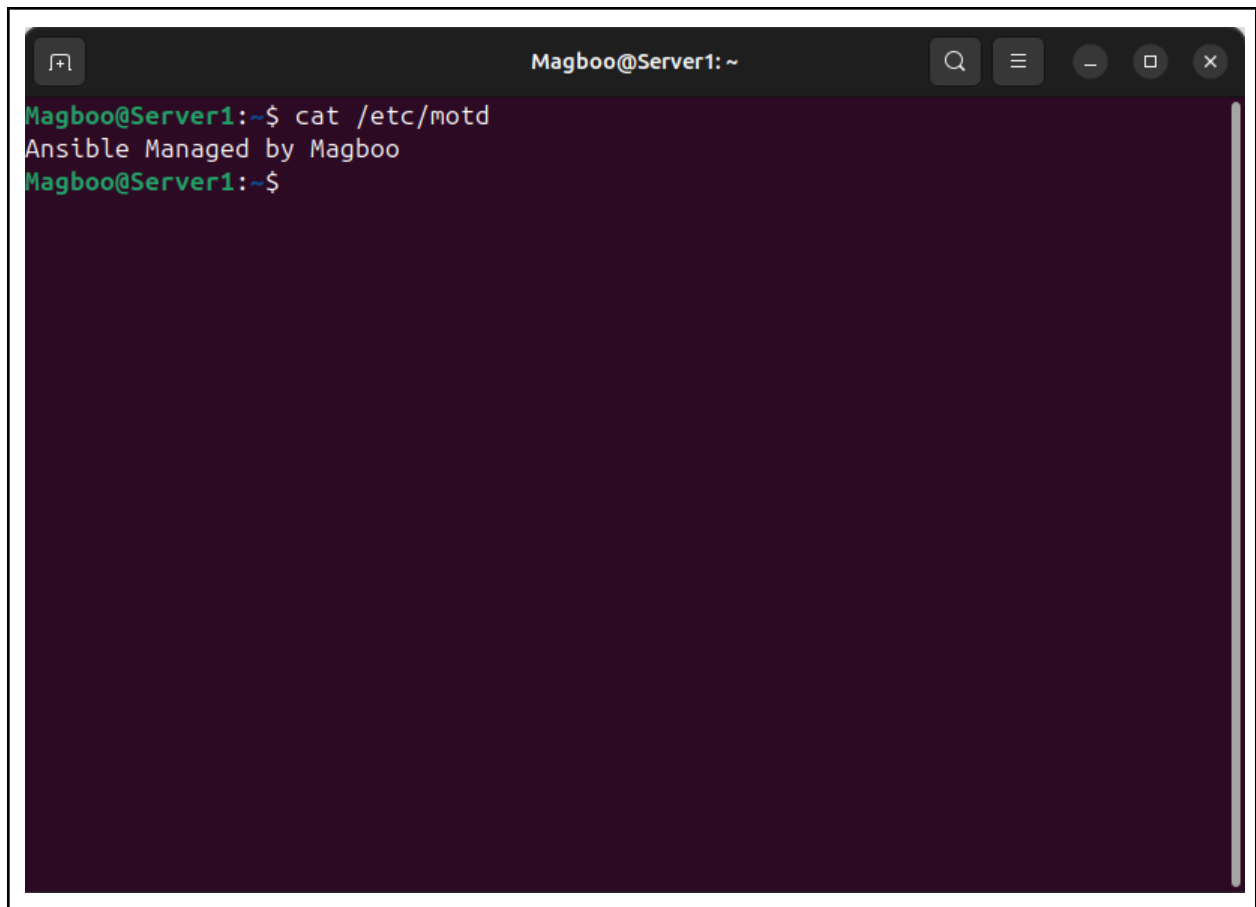
TASK [apache2 : install apache and php for CentOS] *****
skipping: [Server1]
ok: [Server2]

PLAY RECAP *****
Server1      : ok=5    changed=0    unreachable=0    failed=0    skipped=8    rescued=0    ignored=0
Server2      : ok=10   changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

Magboo@LocalMachine:~/Final_Exam_Magboo$

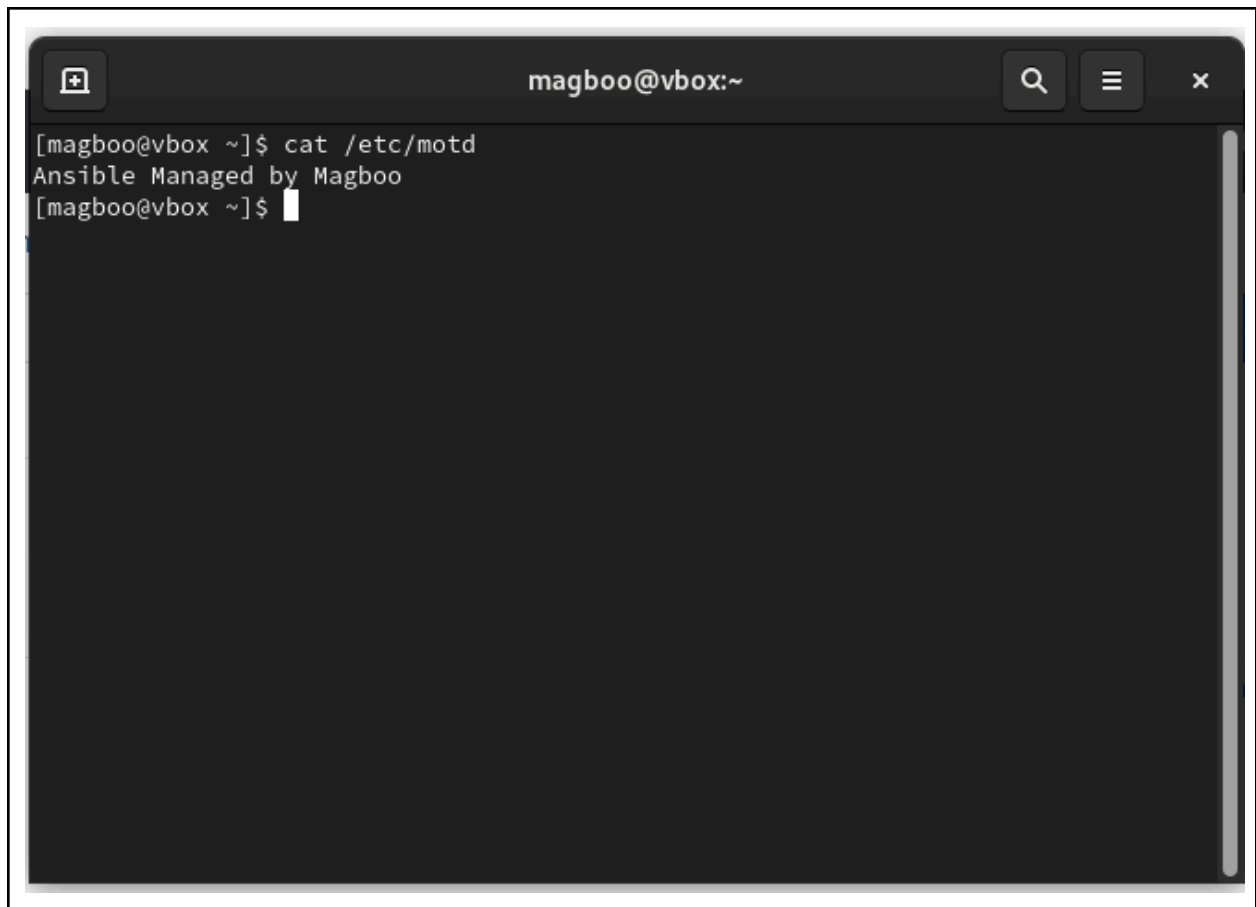
```

4.4 Change Motd as "Ansible Managed by <username>"

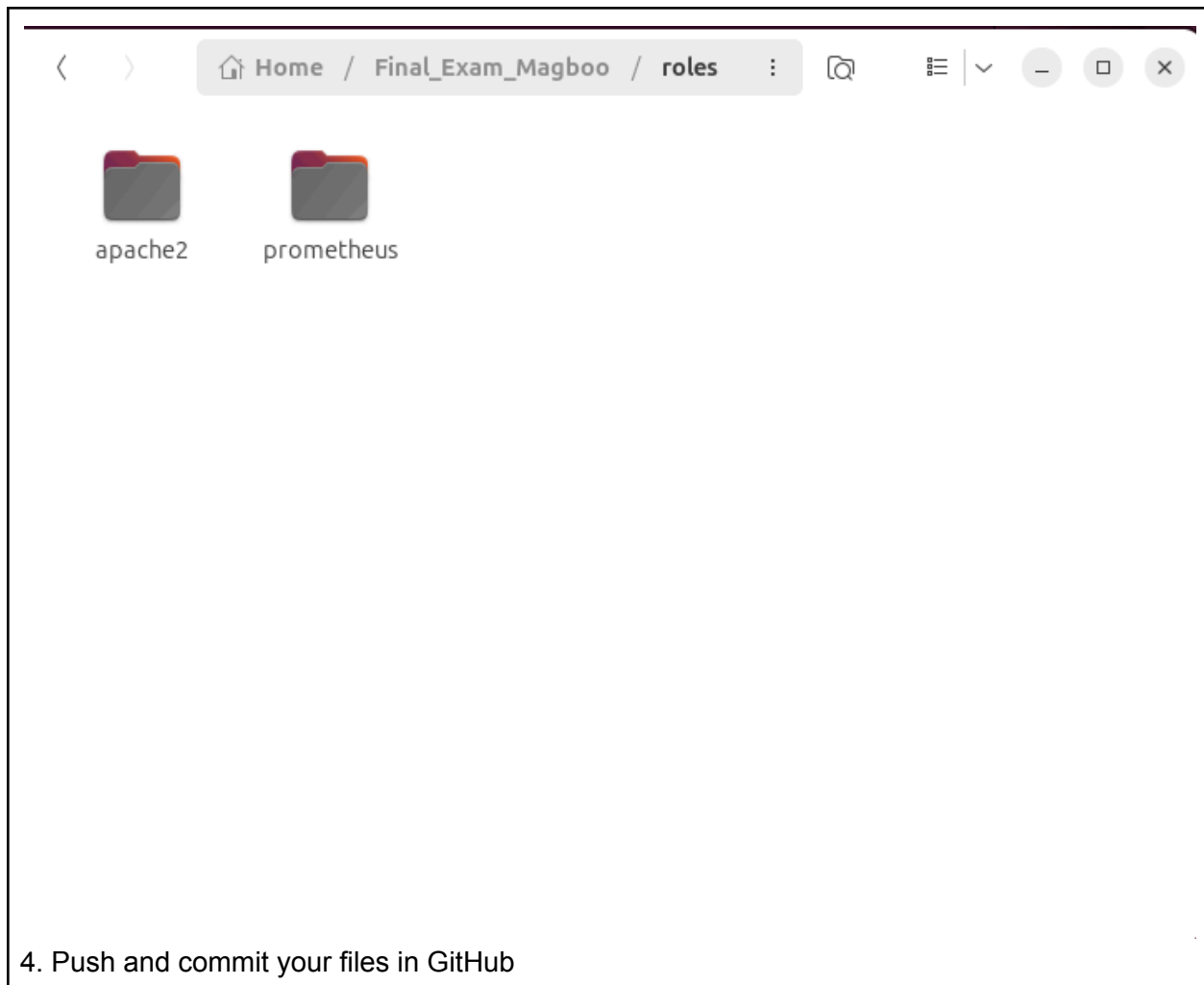


A terminal window titled "Magboo@Server1: ~" with standard window controls (search, menu, zoom, close). The terminal shows a command being executed and its output.

```
Magboo@Server1:~$ cat /etc/motd
Ansible Managed by Magboo
Magboo@Server1:~$
```

A terminal window with a dark background and light text. The window title bar shows 'magboo@vbox:~' and standard window controls. The terminal content shows a command being executed and its output.

```
[magboo@vbox ~]$ cat /etc/motd
Ansible Managed by Magboo
[magboo@vbox ~]$
```



```
CN_Magboo [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Nov 19 14:12
Magboo@LocalMachine: ~/Final_Exam_Magboo

Magboo@LocalMachine:~/Final_Exam_Magboo$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   config.yml
        modified:   inventory.yml
        modified:   roles/prometheus/tasks/main.yml

no changes added to commit (use "git add" and/or "git commit -a")
Magboo@LocalMachine:~/Final_Exam_Magboo$ git add config.yml
Magboo@LocalMachine:~/Final_Exam_Magboo$ git add inventory.yml
Magboo@LocalMachine:~/Final_Exam_Magboo$ git add roles/prometheus/tasks/main.yml
Magboo@LocalMachine:~/Final_Exam_Magboo$ git push origin main
Everything up-to-date
Magboo@LocalMachine:~/Final_Exam_Magboo$ git commit -m "Final Exam"
[main 7681661] Final Exam
 3 files changed, 61 insertions(+), 43 deletions(-)
Magboo@LocalMachine:~/Final_Exam_Magboo$ git push origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 10 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1.27 KiB | 1.27 MiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MagbooMattClemence/Final_Exam_Magboo.git
 b13696c..7681661  main -> main
Magboo@LocalMachine:~/Final_Exam_Magboo$
```

Final_Exam_Magboo Public

[Pin](#)[Watch](#) 0

main 1 Branch 0 Tags

[Add file](#)[Code](#)

MagbooMattClemence Final Exam

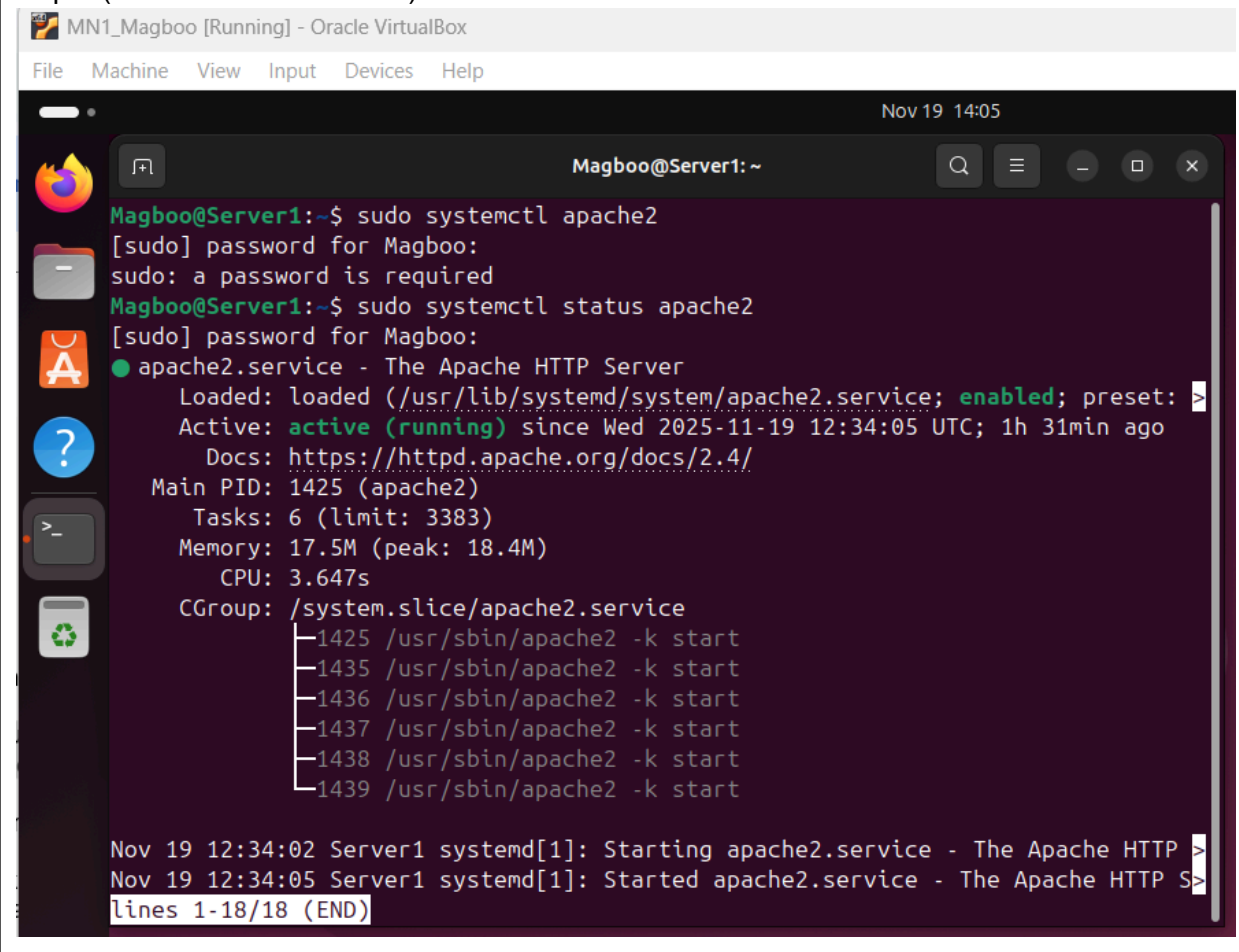
7681661 · now 3 Commits

roles	Final Exam	now
README.md	Initial commit	5 days ago
ansible.cfg	Final exam part 1	5 days ago
config.yml	Final Exam	now
inventory.yml	Final Exam	now

[README](#)

Final_Exam_Magboo

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)



The screenshot shows a terminal window titled "Magboo@Server1: ~" with a search bar and window controls. The terminal output shows the following commands and results:

```
Magboo@Server1:~$ sudo systemctl apache2
[sudo] password for Magboo:
sudo: a password is required
Magboo@Server1:~$ sudo systemctl status apache2
[sudo] password for Magboo:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
   Active: active (running) since Wed 2025-11-19 12:34:05 UTC; 1h 31min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1425 (apache2)
      Tasks: 6 (limit: 3383)
     Memory: 17.5M (peak: 18.4M)
        CPU: 3.647s
     CGroup: /system.slice/apache2.service
            └─1425 /usr/sbin/apache2 -k start
              └─1435 /usr/sbin/apache2 -k start
                └─1436 /usr/sbin/apache2 -k start
                  └─1437 /usr/sbin/apache2 -k start
                    └─1438 /usr/sbin/apache2 -k start
                      └─1439 /usr/sbin/apache2 -k start

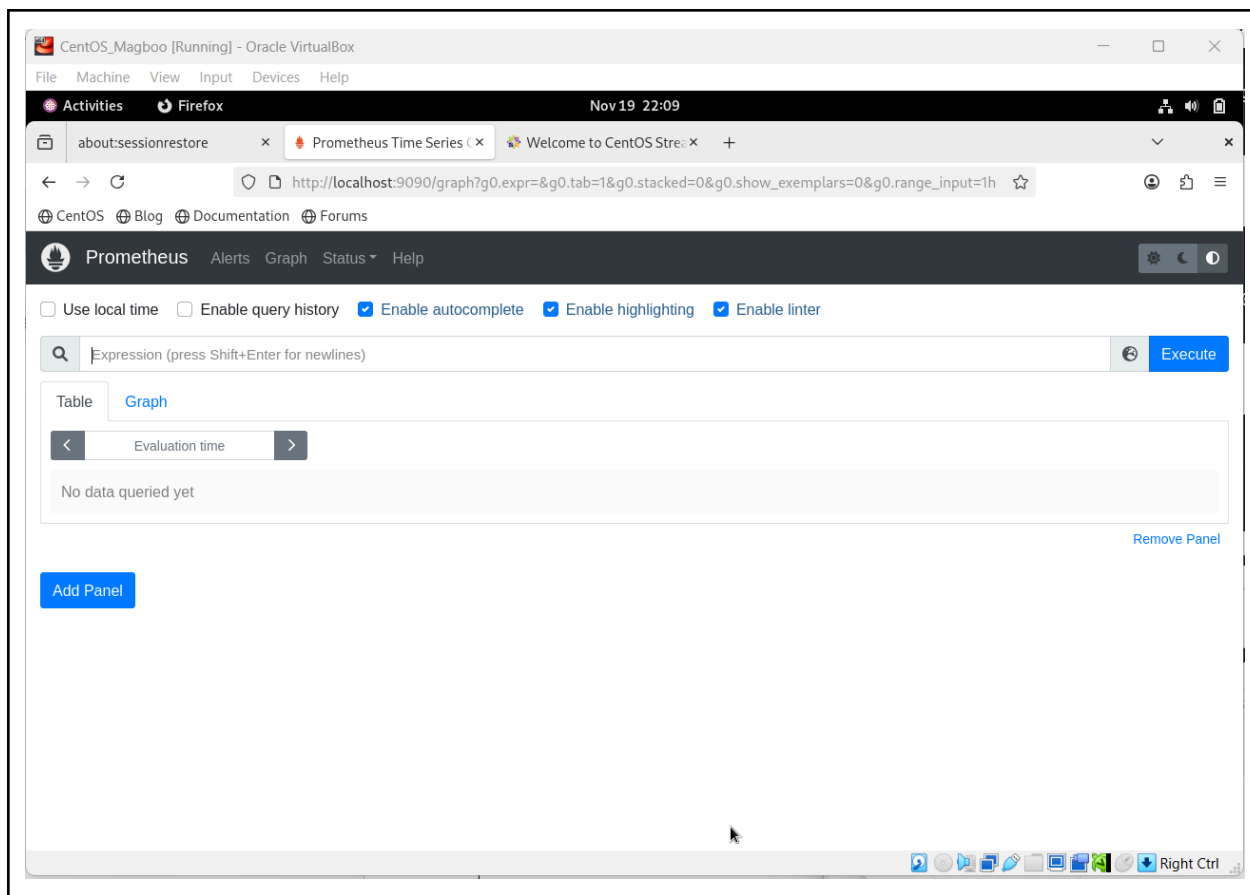
Nov 19 12:34:02 Server1 systemd[1]: Starting apache2.service - The Apache HTTP
Nov 19 12:34:05 Server1 systemd[1]: Started apache2.service - The Apache HTTP S
lines 1-18/18 (END)
```

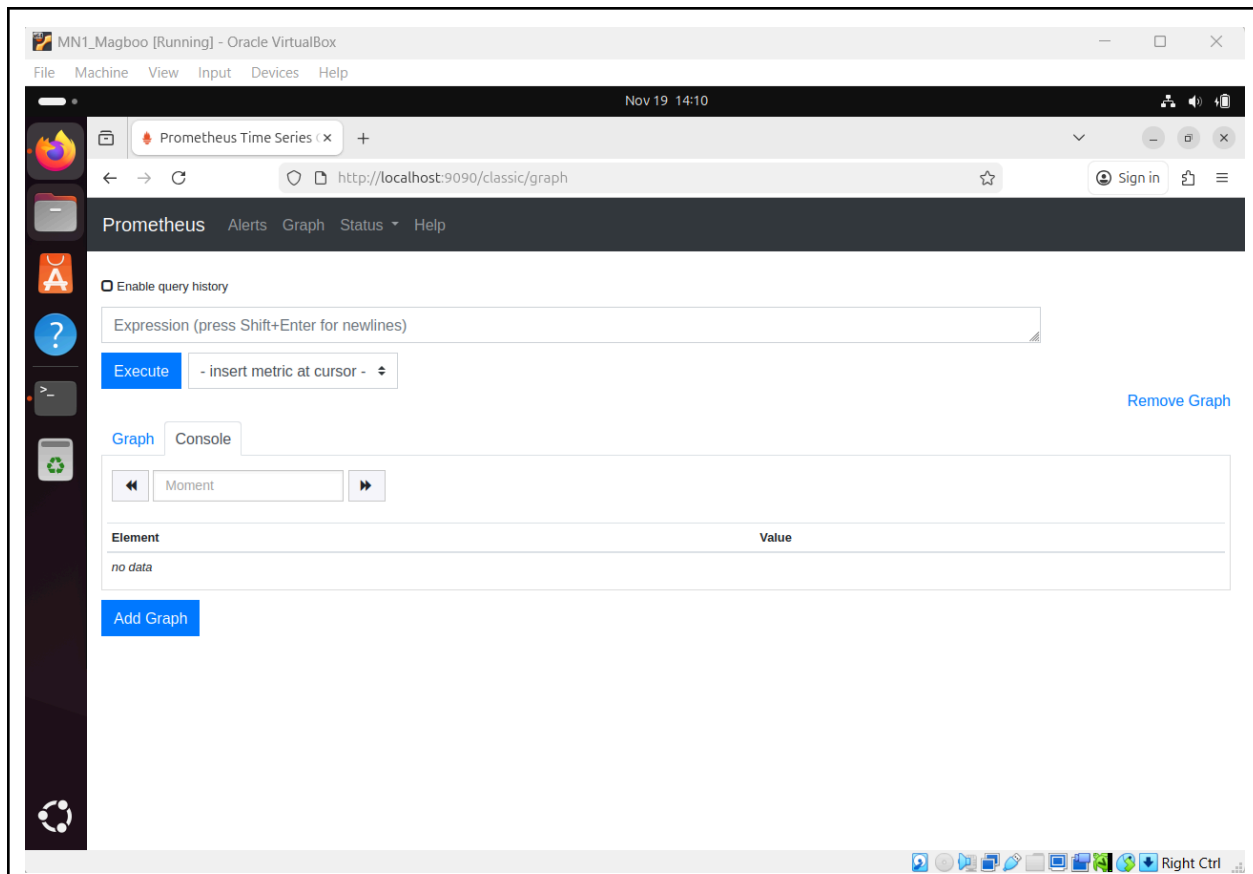
The terminal window is part of an Oracle VM VirtualBox environment, as indicated by the title bar "MN1_Magboo [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The terminal window itself has a title bar "Magboo@Server1: ~" and standard window controls (search, menu, zoom, close). The terminal output shows the successful installation and status of the Apache HTTP Server, including the command to start it, the password prompt, and the status output showing it is active and running.


```
magboo@vbox:~ — sudo systemctl status httpd

[magboo@vbox ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Wed 2025-11-19 20:33:40 PST; 1h 33min ago
     Docs: man:httpd.service(8)
  Main PID: 957 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes"
     Tasks: 177 (limit: 22983)
    Memory: 18.5M (peak: 19.3M)
       CPU: 12.380s
    CGroup: /system.slice/httpd.service
            └─ 957 /usr/sbin/httpd -DFOREGROUND
               1083 /usr/sbin/httpd -DFOREGROUND
               1085 /usr/sbin/httpd -DFOREGROUND
               1087 /usr/sbin/httpd -DFOREGROUND
               1090 /usr/sbin/httpd -DFOREGROUND

Nov 19 20:33:39 localhost.localdomain systemd[1]: Starting The Apache HTTP Serv>
Nov 19 20:33:40 vbox httpd[957]: AH00558: httpd: Could not reliably determine t>
Nov 19 20:33:40 vbox systemd[1]: Started The Apache HTTP Server.
Nov 19 20:33:40 vbox httpd[957]: Server configured, listening on: port 80
lines 1-22/22 (END)
```





5. For your final exam to be counted, please paste your repository link as an answer in this exam.

Github link: [MagbooMattClemence/Final_Exam_Magboo](https://github.com/MagbooMattClemence/Final_Exam_Magboo)

Note: Extra points if you will implement the said services via containerization.